Legend

Inputs *(variable name in the code, to remove in further iterations)*

Intermediary product

Outputs

Snow detection processing

Mathematical formulation

function used (to remove in further iterations)

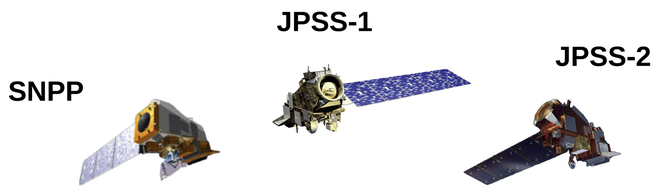
Geometric processing

function used (to remove in further iterations)

Script name

Pipeline

Parameters			
Varia	Name	Value	Units
a	FSC regression	1.45	-
b	FSC regression	-0.01	-
t_{fsc}	Low FSC threshold	0	-
t_{ed}	Low red reflectance	17	pixe
t_{lr}	Low red reflectance	0.07	-
t_{rd}	Color distance	40	%
C_{ma}	Clear observation	1 = probabl	-



VIIRS_SDR

IMG
I3, I1 bands
(SDRviirsI)
Geolocation *(GITCO)*

SnowAge

SnowAge product
R' G' B' Bands
(CCNPPJJSNOW)

MAIA cloud classification

Cloud classification
(*viiCloudTypeImg*)

Snow cover fraction

NPPhdf2img_NDSI.py

Reprojection to L3 grid

pyreseample.kd_tree.resample_nearesi

View zenith angle extraction

NDSI calculation

$$\mathrm{NDSI} = \frac{\mathrm{I1} - \mathrm{I3}}{\mathrm{I1} + \mathrm{I3}}$$

FSC regression

$$\mathrm{FSC} = a\mathrm{NDSI} + b$$

Preliminary FSC

app_imgViirsTchauss.sh

Extraction target footprint

gdalwarp

Snow age to highly likely snow mask

$$\sqrt{(R' - 255)^2 + B'^2 + C'^2} < 255\ t_{rd}$$

(convert -red -fuzz + convert -alpha extract)

Extraction target footprint

gdalwarp

Cloud classification to cloud mask

$$CT < C_{max}$$

GTiffModifColorTable + gdal_translate

Mask Type Color Table
(ct_MaskCloudType)

Granule concatenation

concatViiCTh5.py

Reprojection to L3 grid

maiahdf2img.py

Highly likely snow mask

Cloud mask

putMasksAndColor.py

Preliminary snow detection

$$\mathrm{FSC} > t_{fsc}, \mathrm{FSC} > 1 \Rightarrow \mathrm{FSC} = 1$$

NDSI artifact screen

$$\mathrm{ED} > t_{ed}$$

Euclidean distance transform

$$\mathrm{ED} = \mathscr{EDT}(\text{highly likely snow mask})$$

scipy.ndimage.distance_transform_edt

Low reflectance screen

$$\mathrm{I1} > t_{lr}$$

Water body mask application

Water body mask
(based on corine land cover v2006)

Cloud mask application

Daily snow cover products

Daily view zenith angle products

make_snow_cover_compsite.py

View zenith angle composition algorithm

Météo-France VIIRS multi-platform composite